

Boğaziçi University
ME492 Project Report

ROBOCOT: COTTON HARVESTING ROBOT

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EXECUTIVE SUMMARY

This report outlines RoboCot, a simulation program built for a lab prototype that handles autonomous cotton boll detection and picking with a rover based dynamic robotic manipulator. A full “end to end” perception to action pipeline is demonstrated. Inside a realistic mock field simulation, cotton bolls are detected and their target positions are estimated in 3D using depth sensing. From there, safe approach motions are planned and repeatable pick and place actions are executed to drop the cotton into a compact collection bin. We define this system as a technology demonstrator instead of a fully field ready machine. Finally, the system's performance limits regarding occlusion, clutter, and shifting lighting are challenged and investigated within the simulation environment to pave the way for bigger, future projects

We defined a complete system concept, operating steps and engineering requirements so the prototype could be built and tested. However, after encountering problems with the robotic manipulator, the project was fully transformed into a detailed computer simulation. This simulation mimics every intended challenge for the real life prototype and it is expected to act as a guideline for future students to implement a physical version using a suitable robotic manipulator. The intended workflow includes detecting cotton clusters, estimating targets in three dimensions and picking individual bolls. This picking step features a basic ripeness check performed by the system's vision based machine learning algorithm, followed by pick confirmation and transfer to a collection bin. Design choices are guided by the “Product Design Specifications” which cover subjects like quality, maintainability, size, weight, safety, ergonomics, reliability, environmental conditions and cost. Key constraints involve an operational reach of 1300 mm, operation across an expanded mock field area utilizing the differential drive mobile platform, and a collection bin size of 15 cm x 15 cm x 15 cm. Finally, a parallel gripper stroke of exactly 50 mm is required, featuring an adjustable pinch force between 20 N and 185 N to ensure delicate handling of the plant. Given the robust industrial kinematics of the manipulator, tip deflection under the nominal end effector payload is negligible.

The simulation is built around mimicking the conditions of the real life prototype. A modular architecture and plan are used to support simpler testing and debugging processes afterward. Cotton bolls are detected using a machine learning model. Camera focus calibration is applied to the pixel data and depth information is embedded; these two steps are then used to convert the detections into three dimensional coordinates for the desired target points. A standard motion sequence is then executed: approach, grasp, lift and place into the bin. Safety is addressed through speed limits and hardware controls. For instance, the gripper tip speed is limited to Tool speed is capped at 0.3 m/s near humans for collaborative safety, though the manipulator can execute trajectories up to 1 m/s. An emergency stop is provided to cut actuator power. Maintenance is supported through easy lens access for the RealSense D435 depth camera, as well as remote software or model updates handled by the onboard computer via the ROS system. Measurable targets are defined so that success and failure modes can be reported clearly within the MoveIt2 and Gazebo simulation environments. With the transition to an industrial manipulator, gripper tip positioning accuracy and pick position repeatability are significantly tighter than the previous ± 5 mm and ± 3 mm specifications, ensuring highly reliable vision guided localization. Cotton detection accuracy is targeted at 90% or higher under conditions

that mimic the actual field within the mock setup. A staged validation approach was recommended. First, sensor compute control integration and calibration stability should be verified in the mock field. Next, perception accuracy and three dimensional targeting errors

should be measured. Finally pick success rate, cycle time and dominant failure causes such as lighting effects and reach limits must be evaluated. Through this approach, reusable outputs are produced, including a labeled dataset.

1. INTRODUCTION

1.1 Background and Motivation

Agriculture is under pressure to produce more food while also meeting sustainability goals and dealing with climate related changes [1]. At the same time, a “digital revolution” is expected to emerge in the agriculture and food sector. This change is linked to the wider use of sensors, software tools and farm management systems [1]. It is expected that data and robotics will increasingly influence both the decision making process and the physical work on farms. However, technology adoption can stay limited if farmers are not shown the clear value of this implementation [1]. Issues such as data ownership, system compatibility, training needs and high investment costs are often mentioned as barriers to transferring this technology revolution into the agriculture and food sector [1].

Many important farming tasks are still labor-intensive today, and access to human labor can be unstable since humans are not as reliable as automated systems [3]. Agricultural work has been described as strongly dependent on manual labor and highly sensitive to disruptions coming from the human factor [3]. Also, long term demographic and economic changes can worsen labor shortages, since fewer younger workers may prefer physically demanding jobs in today's global conditions [3]. On a wider scale in the big picture for the future, robotics and autonomous systems are also highlighted as major global technology trends, meaning automation pressure is expected to grow across many industries, including the agricultural industry [2].

Within agricultural robotics, harvesting is considered one of the hardest application fields [3][4]. Perception, motion planning and delicate manipulation must be combined in an unstructured environment, as Figure 1 shows, where plants are not fully rigid, targets are partially hidden by leaves and other herbal obstacles and lighting changes [3]. Planting and harvesting have been described as seasonal operations that require much heavy effort and many automation concepts have been reported using RGB cameras, three dimensional sensing, laser scanners, manipulators and grippers [4]. Interest in autonomous and semi-autonomous systems has often been linked to reducing the top high workload tasks, especially harvesting [4]. Robotics roadmaps for the food industry also suggest that automation becomes most attractive when manual labor is high cost, raw materials vary widely and speed is limited by human factors [5].



Figure 1: Cotton Row With Ripe Cotton Bolls

Even though progress has been made, important technical problems still remain in real harvesting conditions [3]. Obstacles and changes in ambient lighting are repeatedly reported as major challenges [3]. Harvesting performance is often inconsistent across different systems and scenarios depending on these major challenges [3]. Reported success rates and cycle times can vary widely, which shows that results depend strongly on the environment of the subject and the system design [3]. This indicates that robust harvesting is limited not only by mechanical hardware but also by perception quality, calibration stability and the reliability of the perception-to-motion integration in cluttered environments [3].

For these reasons, a controlled laboratory demonstrator is strongly justified before moving toward field deployment [3]. A technology demonstrator allows key limitations to be reproduced and measured using clear test procedures [3]. This is especially true for robust vision under occlusion and lighting changes, as well as safe manipulation in clutter [3]. Cotton boll picking as an agricultural activity fits this purpose well [3]. Targets can be visually identified for learning-based detection, while realistic difficulties such as clusters, leaf and branch obstacles, and grasp slips can still be introduced in a repeatable mock field setup [3]. As a result, limits can be reported clearly, and reusable outputs such as interfaces, calibration procedures, evaluation metrics, and datasets can be produced for future improvements [3].

1.2 Problem Definition & Project Significance

Cotton is a very important natural fiber, mostly because we use it so much in clothing and household goods [6]. Picking cotton essentially means carefully taking the open bolls off the plant. You have to keep the fiber clean and avoid grabbing extra plant parts like leaves or stems. This is actually pretty tricky since not all bolls open at the same time. As you can see in Figure 2 a fully open boll ready for picking might sit right next to one that is completely closed or only half open [7].



Figure 2: Closed Cotton Boll Next to Fully Open Boll

Currently, farmers usually just harvest everything once at the end of the season. They also plan leaf shedding (defoliation) to help the big harvesting machines work better and protect the lint quality [7][10]. But doing things this way has some downsides. For instance, bolls that open early are left out in the bad weather until the final harvest day, which can really lower the fiber's quality [7]. Plus, standard harvesting machines are massive. Their heavy weight causes soil compaction, which is a big worry for the future of farming [7]. On top of that, comparing

different harvest methods reveals other problems. When cotton is harvested with a stripper machine, it ends up with more unwanted plant bits and needs a lot more cleaning than cotton picked with a spindle [8][9].

This is why robotic cotton harvesting is so important. Robots can be built as much smaller and lighter platforms. They also allow for selective picking. This means they can collect a boll right after it opens instead of waiting for the whole field to be ready at the end of the season, which helps to keep the fiber quality high [7]. Even so, building these robots is technically very hard. Developing agricultural robots is notoriously difficult because they have to handle motion planning, perception, and physical manipulation all at once. Plus, they have to do this in messy outdoor environments with changing lights and random obstacles [3][4]. We see these exact struggles in specific cotton studies too. For example, one test on a robotic cotton picker showed it only managed to grab about 66% to 85% of the cotton from a boll. The time it took to pick also varied a lot, taking anywhere from a few seconds to over ten seconds based on how it was controlled [7]. Because of these challenges, it makes a lot of sense to build a smaller, lab scale prototype before putting a robot directly into a real farm. Working in a lab lets researchers recreate and measure the main problems mentioned in robotics studies under controlled conditions. They can safely test things like how well the camera sees through obstacles and changing light or how smoothly the robot turns what it sees into physical movement [3][4]. Taking this step also fits perfectly with current plans for digital farming, which focus heavily on robotics, data tools and solving user training problems [1]. It also matches up with general workforce reports showing that automation is a huge trend across all sectors [2]. Overall, cotton is a great subject for these kinds of lab experiments. Previous research has already shown that cotton bolls can be tracked using 3D depth cameras even in harsh sunlight [11]. Furthermore, modern machine learning systems are specially built to deal with tricky backgrounds and changing field lighting [12].

1.3 Benchmarking

Today, farmers mostly use two kinds of machines to harvest cotton: the spindle picker (Figure 3) and the brush roll stripper [13]. People consider the spindle picker a selective machine because it mainly grabs the fully open bolls, leaving most of the unwanted plant parts behind [13]. On the other hand, the brush roll stripper is not selective at all. It just tears off everything at once like good cotton, unripe bolls, sticks and leaves [13].



Figure 3: Spindle Picker Cotton Harvester

This creates a big difference when we look at the amount of trash mixed in and the overall fiber quality. Since strippers grab so much extra plant material, a lot more junk goes into the "ginning" stage (where the fibers are separated). Naturally, this means more cleaning is needed, which can slow down processing and make the final lint look less neat [9][13]. The numbers really show this gap, even if farmers use field cleaners. Cotton from a spindle picker usually has about 68 to 70 kg of trash per bale, while stripper harvested cotton carries around 170 to 173 kg of trash per bale [13]. This extra trash also lowers the fiber quality, mostly because the stripper collects a lot more unripe fibers compared to the picker [13]. There are also trade offs when it comes to cost and daily use. Strippers are simpler machines, making them cheaper to buy and run. Spindle pickers however are a lot more complicated and need more daily maintenance just to keep their performance steady [13]. Because both of these machines are made to collect huge amounts of crop at the very end of the season, picking selectively isn't really their main goal. Instead, the final quality of the cotton depends heavily on how well the machine is set up, along with the general condition of the crop [13].

How farmers prepare the fields beforehand makes a huge difference too. To speed things up and stop leaves from dirtying the collected cotton, farmers often use defoliation (chemicals that make leaves fall off) and try to pick at the perfect time [10]. But even with all that careful prep work, the same old problems stick around. Things like bad weather in the field and the exact type of machine used will still change the amount of unwanted trash and the final fiber quality [13]. Beyond the machines themselves, their massive size causes real problems in the field. Driving these heavy harvesters back and forth really packs the soil down, especially if the ground is wet [14]. In fact, field tests show that after a cotton picker drives over the dirt, the soil becomes noticeably harder up to 0.6 meters deep (as seen in Figure 4). This compaction happens with both the older basket style pickers and the newer, much heavier machines that pack cotton into round modules [14].



Figure 4: Soil Compaction Caused by the Harvesters

All of these issues explain why the farming world is pushing so hard for robotic alternatives. Sure, traditional machines are great for picking tons of cotton fast. But they still mix too much dirt into the crop, get messed up by the weather and ruin the ground with their extreme weight. Because of this, researchers really want to create lighter robots that can pick selectively. It seems like it won't be easy though. Harvesting is still known as one of the hardest jobs you can

give a robot. It struggles with leaves blocking the camera's view (occlusion), the sunlight constantly changing and the physical challenge of navigating a robotic arm through a messy, crowded plant [3][4].

1.4 Literature Review

Research into farming robots has been conducted for decades, yet the actual harvesting of a crop is still considered one of the most challenging activities to automate [3][15]. While simpler tasks like field mapping or crop monitoring are easily handled, harvesting requires a living plant to be physically interacted with and a target to be accurately seen, rather than a human worker simply being assisted. Today, a harvesting robot is viewed by most engineers as a complete "perception to action" chain where every single component must be perfectly connected to the next. First, the target is detected and its position is estimated in three dimensions. After that, a safe path is planned so the product can be extracted without the harvested material being damaged [3][4]. Because these steps are so deeply interconnected, harvesting is frequently described by experts as a continuous pipeline from perception to action. Therefore, it is considered much more complicated than a material simply being grasped from nature [3].

When the construction of most of these robots is examined, a similar blueprint is generally followed. Operations are usually begun with "vision" where regular cameras are mixed with depth sensors so the working environment can be observed [4][15]. Once a picture is captured, machine learning is utilized so the target harvest material, such as a fruit or a cotton boll, can be located [3]. These smart models are highly favored for outdoor work because they are not easily confused by complicated backgrounds or shifting sunlight throughout the day [3][12]. These factors are critical for outdoor environments, as outputs can be significantly affected by such obstacles. After the target is found, a map of 3D coordinates is created using sensor data. Precise movement coordinates are then sent to the robotic arm so the crop can be pulled off the plant by a gripper or suction tool [3][15][16]. While a general design idea is provided by this, gripper (shown in Figure 5) and suction tools are separated by their efficiency. Since more extra material may be pulled into the harvest by suction tools, grippers are generally desired so fewer unwanted materials are collected alongside the cotton bolls [16].



Figure 5: Harvester Gripper Prototype Designs

The primary problem is that consistent performance is not always achieved by these robots [3][15]. Spaces where designed systems work continuously without random obstacles being encountered can be created in lab environments. However, while perfect operation might be observed in a clean lab, the success rate is often dropped the moment the machine is placed in

a real field. This occurs because farm environments are characterized by chaos. They are not organized like a factory; plants are blown around by the wind, the view is blocked by branches and targets are often hidden under plant leaves [3]. These small environmental issues are accumulated and bigger problems are created for the working principle of the harvesting systems. Because the processes are so interconnected, a massive mistake in positioning can be caused by a tiny error in depth sensing. Consequently, branches might be hit, targets might be missed or grips might be lost, by which the product or the harvesting system itself can be damaged an outcome that is highly undesirable [3]. This job is made incredibly difficult by a few specific bottlenecks. The most significant issue is defined as blockages, where the target is hidden by leaves and branches [3][15]. Outdoor lighting is also faced, where sensors can be easily tricked and accuracy can be ruined by shadows and glare [3]. Additionally, since plants are not fully rigid objects like metal, unpredictable movements can be caused across the whole branch just by a simple touch by which the coordination systems of the harvester may be disrupted [3][15]. Finally, the risk of damage must be addressed. The harvested material must be removed without the product being destroyed or the plant which serves as the source for future harvests being hurt. This requirement makes the design of the gripper incredibly important [15]. Lastly the camera and the robot arm being kept perfectly synchronized is experienced as a constant struggle. In a rough farm environment, the system's alignment can be caused to drift by vibrations, making the robot's movements inaccurate [18]. When a robot is failed, a single big mistake is rarely blamed rather a snowball effect of small errors is usually observed. For example, if a cotton boll is partly hidden by a leaf, an inaccurate depth reading might be received by the robot, causing the wrong spot to be grabbed. As a result, an unsuccessful operation is executed and overall efficiency is dropped [3][18]. This is especially true for cotton, where the bolls are frequently tangled and hidden. These factors clearly support the need for a controlled demonstrator. Through this demonstrator, perception stability, 3D localization errors and pick success rates are allowed to be measured in a repeatable setup before real field conditions are targeted for a larger scoped project [3][18]. In modern agriculture, one of two major roles is taken on by autonomous systems [19]. They are utilized either as data acquisition systems for monitoring or as operational action tools for harvesting. Unmanned Aerial Vehicles (UAVs/drones), as seen in Figure 6, are mainly served as the "eyes" of these complex systems. Equipped with RGB lenses, multispectral cameras, and thermal sensors, fields are mapped from the sky so water stress, nutrient deficiencies or spread diseases can be identified. A precise "mapping to analysis" workflow is enabled by this data, ensuring that interventions are not made randomly but are exactly targeted only where needed. For physical operations, the lead is taken by Unmanned Ground Vehicles (UGVs) and autonomous platforms, as their specific capabilities are better fitted for harvesting actions. While drones are excellently utilized for rapid spraying in difficult terrains (like steep orchards), limitations are faced regarding payload capabilities and energy issues when common field crops like wheat, corn or cotton are harvested. High outputs and heavy duty mechanisms are required by these crops, making ground based robots a far more practical choice for the job [19].

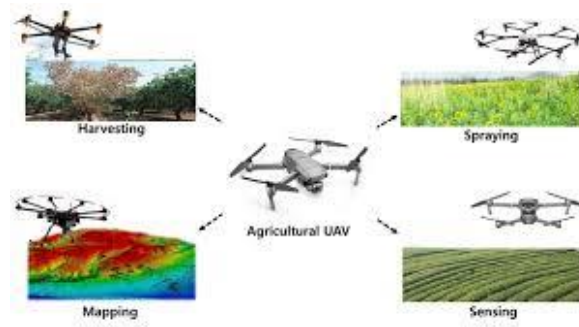


Figure 6: Unmanned Aerial Vehicles in Agricultural Use

Creative solutions are being pioneered by companies like Tevel, where "Flying Autonomous Robots" (Figure 7) are deployed. In this system, fruit is picked using robotic arms while the drone remains connected to a ground power source [20]. However for large scale massproduced crops such as wheat and cotton, autonomous ground platforms are heavily relied upon by the industry. Fully autonomous tractors and harvesters have been introduced by John Deere, where fields are navigated with centimeter level high precision using computer vision and GPS [21]. Other innovations are also observed, including SwarmFarm, where fleets of smaller autonomous ground chassis (seen in Figure 8) are utilized. These serve as moving platforms so tasks like weeding and harvesting support can be performed without the soil being compacted, thanks to the distributed and low pressure of the vehicle [22]. Similarly, a UGV platform is used by Carbon Robotics, where high power lasers are equipped so weeds can be eliminated autonomously [23].



Figure 7: Tevel Flying Autonomous Robots™



Figure 8: "SwarmBot 5" Sprayer

1.5 Theoretical Background

The construction of a cotton picking robot is driven by a "perception to action" system which is entirely transferred into the simulation of the RoboCot, where sensor data is translated into physical movement through a structured software framework [3][4][15]. Several key areas are included in this process such as agricultural automation, robot simulation, motion planning, object detection and 3D perception. To ensure effective communication between the perception and control modules, these components are usually linked together by a middleware program like ROS 2 [24].

Agricultural Automation and Detection Challenges

Harvesting is considered a major focus within precision agriculture because unstructured environments must be interacted with by the robot [3][4]. Cotton detection is made particularly difficult since the bolls are frequently grown in clusters, hidden by leaves or affected by unpredictable and shifting outdoor light conditions [3]. These problems are solved by utilizing learning based detection, as complicated backgrounds and lighting changes are handled much better by these models than by traditional rule based programming [3][12].

Motion Planning and Control

A mathematical model is required by the manipulator inside the simulation so its joint angles can be linked to its hand position. The location of the hand based on the joints is calculated by forward kinematics, where the required joint angles to reach a specific target are calculated by inverse kinematics [25]. Within ROS 2, "MoveIt 2," as seen in Figure 9, is utilized as the standard tool so these movements can be planned while obstacles are avoided [25]. On the hardware side, motor commands and sensor readings are managed in a modular way through the use of "ros2_control" [24].

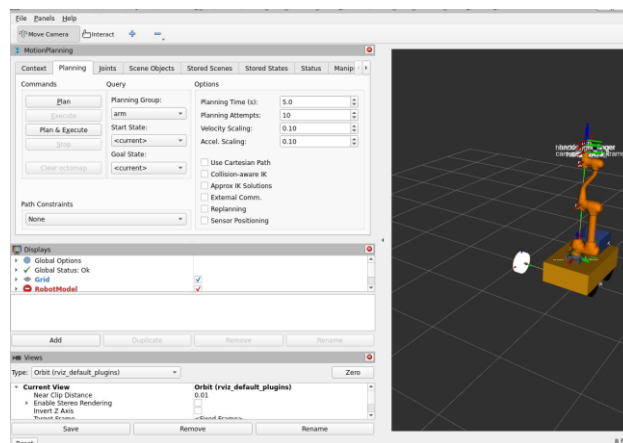


Figure 9: "Moveit2" for Inverse Kinematics

Robot Simulation and Digital Twins

First of all robot behavior is validated in a virtual environment so the risk of accidents can be reduced and potential problems can be seen beforehand. Initial reason was for tests to be conducted before being conducted on real hardware but as the project scope shifted from real world to a full simulation environment, these tests become the tests for the simulation itself. The physical structure and joints of the robot are described using the Unified Robot Description Format (URDF) so a "digital twin" can be created [27]. This digital twin model is then placed

into a physics based simulator Gazebo (seen in Figure 10), which is often run alongside ROS 2 [26][27]. Whether the robot can reach its targets is checked and potential problems are spotted through this useful simulation [3][26].



Figure 10: Simulation Environment in Gazebo

Object Detection and Tracking

Cotton perception is handled through object detection, where each boll is identified by a bounding box and a confidence score [12]. YOLO style detectors are run on live video feeds by utilizing the Ultralytics library [28]. Before cotton can be recognized by the robot, datasets must be labeled using tools like "Roboflow Annotate" [29]. Consistent labeling is considered crucial because unwanted low performance in the actual field can be caused by errors in the dataset [12]. As the robotic arm is moved, cotton bolls are kept track of by multi object trackers such as ByteTrack or BoT-SORT [28]. A unique ID is assigned to each boll by these tools, by which harvested materials are counted and picking the same target twice is avoided [31][32].

3D Perception and Coordinate Transforms

A 3D location, found by using a depth or stereo camera, is required for the picking process. Depth (Z) can be estimated by the formula $Z \approx \frac{f \cdot B}{d}$, where " f " is defined as the focal length, " B " is defined as the distance between lenses and " d " is defined as the disparity [15]. Once a target is found by the camera, its position must be converted into the base coordinates of the system. In ROS 2, this conversion is managed by the TF2 library, by which the relationship between different parts of the manipulator is tracked [24]. If movement is adjusted based on live camera feedback, the process is called "visual servoing." Targets that might be moved or partially blocked by branches are which is considered the general casemore easily reached when visual servoing is utilized [3][33].

3D Model Extraction (Optional Extension)

For more advanced analysis, frameworks like SAM3D can be used so 3D masks of the scene can be created without extra training being needed [34]. Through this method, the entire structure of the plant and the cotton regions can be modeled by the system. Although assets for digital twins or better planning are helped to be created by these tools from Meta and other researchers, they are still considered optional extensions beyond basic picking tasks [35][36].

Industry Context and the Project Gap

Currently, the agricultural cotton harvesting industry is mostly dominated by large scale mechanical systems, which are used at the end of the season when all cotton bolls are fully ripened and are considered harvestable. However, these traditional methods are often limited by the presence of foreign matter, varying field conditions and strict timing constraints, by which overall fiber quality can be impacted [9][13]. Because perfect coordination of robust perception, stable calibration and safe movement under difficult conditions such as lighting changes and visual obstructions is required, autonomous harvesting is frequently described as a major challenge in robotics [3][4][15][18]. How easily the boll can be removed is heavily affected by the fibrous nature of the cotton. Consequently, the motion and control of the end effector (the gripper) must be tested through physical experiments so efficient picking can be ensured [7]. After the literature is examined, a measurable and repeatable demonstration of a small scale, vision guided cotton picking system for precision agriculture is seen to remain as a significant gap in existing research [1][2][3].

To address this gap, the project is structured around specific and measurable objectives by building a mock field within a controlled laboratory environment. A detection and tracking pipeline is developed using YOLO based models so unique identification for each cotton boll can be maintained across different video frames [12]. 3D target localization is planned to be achieved through RGB sensing. Through this method, perfect alignment between the camera and the robotic arm is ensured by precise coordinate transformations [3][18]. The physical execution of the task is focused on a repeatable sequence of actions: approach, grasp, lift and place. While this is done, both cycle times and success rates are recorded transparently so important metrics can be established [15]. Finally, the entire system is evaluated by utilizing these performance metrics, such as detection accuracy, positioning error and common failure causes, so a solid foundation for future technical improvements can be provided [3][15].

2. DESIGN PROCESS

2.1 Design Criteria and Product Design Specifications:

Product design specifications established in ME429 in the first semester were fully transitioned to ME492's project scope. These design specifications were fully followed while the conduction and continuation of the manufacturing phase of the project. While the project scope was transitioned to a simulation rather than real world manufacturing due to the circumstances the design specifications remained as main conductors of the project flow.

Maintenance

Easy maintenance features like cleaning the camera lens, quick replacement of gripper if needed, simple calibration reduces downtime and supports fast iteration.

MA-01 – The robotic arm shall use modular components so that worn parts (gripper pads, motors, joints) can be replaced individually without replacing the whole arm.
MA-02 – The camera lens shall be directly reachable for cleaning with a cloth or brush without disassembling the camera mount or disturbing its calibration.
MA-03 – Camera recalibration in the mock field shall take ≤ 10 minutes when required.
MA-04 – The system shall support remote software and ML model updates (retraining or new model deployment) via Jetson (over SSH/USB), without physical access to the Jetson board itself.

Size Restrictions

All functionality depends on physically fitting and operating inside the mock field boundaries. Workspace constraints also drive arm selection and camera placement.

SR-01– The manipulator shall provide a maximum operational reach of 1300 mm from the arm base center to the gripper tip in the horizontal direction.

SR-02– The RoboCot system shall be designed to operate within the mock field, which is now scalable beyond the initial 1.0 m length and 0.55 m height due to the mobile base tracking, where the upper 0.25 m of the height is filled with cotton plants.

SR-03– The cotton collection bin internal dimensions shall be 15 cm x 15 cm x 15 cm (± 5 mm).

SR-04– The gripper jaws shall be able to open between 50 - 60 mm to grasp a cotton boll cluster at a single position.

Weight

Weight matters mainly through arm dynamics and tip deflection and is less critical than safety/performance in this stationary prototype.

WT-01– Arm links shall use lightweight materials (aluminium, composite or plastic) so that the total arm mass does not exceed 2.5 kg.

WT-02– The cotton container shall be designed to safely hold about 300 g of cotton without structural failure or excessive deformation.

WT-03– The nominal payload at the gripper (camera ≈ 200 g + cotton boll $\approx$ negligible) shall not cause tip deflection greater than 10mm at full reach.

Safety

Safety is the highest priority because the prototype has moving joints, pinch points at the gripper and powered electronics. Any unsafe motion or unstable behaviour can stop testing and create risk during the demo.

SF-01– All electrical components (Jetson, camera, motor drivers, wiring) shall have basic environmental protection equivalent to minimum IP54 for dust and splash resistance (for outdoor field tests).

SF-02– Arm joint speeds shall be limited so that the gripper tip linear speed ≤ 0.3 m/s when operating near humans.

SF-03– The control software shall handle kinematic singularities by avoiding or slowing motion near singular configurations and preventing unstable joint velocities.

SF-04– The system shall include an emergency stop button that immediately cuts power to the arm actuators while keeping the system in a safe state.

Ergonomics

The system will be operated and monitored by people during the demo. Simple control (GUI), clear status indicators and fast bin handling reduce user mistakes and improve demo

ER-01– A control & monitoring panel (GUI) shall allow the operator to monitor the system, start, stop and pause the system within one or two actions.

ER-02– System status and warnings shall be visible through clear visual indicators (LEDs).

ER-03– The reservoir/cotton bin shall be designed for easy loading and unloading, requiring no tools and < 15 seconds to remove and reinsert.

Aesthetic

Aesthetics is intentionally not a decision maker in this phase. However, basic neatness (cable routing, clean assembly) is still useful to reduce failures and improve usability.

AE-01– Cables shall be neatly routed and secured, with no loose wires hanging into the mock field or moving mechanism range.

AE-02– Major components (arm, bin) should follow a consistent material and color scheme to appear as a unified product.

AE-03– The arm motion should appear smooth and continuous without visible jerks in normal operation (acceleration profiles tuned).

Quality & Performance

The system is only valuable if it can consistently detect cotton and execute the manipulation which is the pick. Vision accuracy, arm positioning accuracy, repeatability and a reasonable cycle time directly define whether the demo is realistic or not and whether the approach can scale later for real life application in a bigger scoped project.

QL-01– The arm shall achieve a positioning accuracy of $\leq \pm 5$ mm at the gripper tip within the working volume.

QL-02– Repeatability of the pick position shall be $\leq \pm 3$ mm over 50 repeated picks at the same target.

QL-03– The ML cotton detection system shall achieve $\geq 90\%$ detection accuracy in typical field-like lighting and dust conditions in the mock setup.

QL-04 – The end to end pick cycle time (from first detection to boll stored in the bin) shall be ≤ 60 seconds under standard conditions.

QL-05– Functional consistency, the system shall maintain an overall successful pick rate $\geq 90\%$ during continuous operation over one test day.

Material Cost

Material cost affects design choices for the arm structure, gripper model and sensor and computing selections.

MC-01– The robotic arm & ml hardware cost (motors, gears, links, gripper, nvidia jetson) and The vision system cost shall be compatible with ML requirements (resolution, fps, dynamic range) shall not be monetary as the project scope shifted to a full simulation environment but they were fully mimicked in this virtual environment in Gazebo.

MC-02– Structural material costs for the platform, reservoir and mounting (aluminum profiles, plastic sheets.) should be minimized while still meeting stiffness and durability requirements.

Manufacturing Cost

The design must be achievable with student manufacturing tools and time. Meaning 3D printable parts, simple assemblies and modular software components that can be reused.

MF-01– Gripper parts shall be 3D printable with standard FDM printers.

MF-02– The full mechanical and electronics assembly (arm, bin, mounts, wiring) should be possible in $\leq 2 - 3$ working day by a 2 person student team.

MF-03– Software development work (ML training, ROS2 integration) should be modular, allowing reuse in future prototypes without large rewrites.

Reliability

The product is a pipeline (camera - detection - position - motion - actuation). If any part fails frequently the full system fails. Reliability is essential for continuous operation during the demonstration.

RL-01– The system shall operate reliably under dust, heat and moderate humidity similar to typical cotton fields in Türkiye (ambient 20°C – 45°C, RH 30–80%).

2.2 Overview of Possible Solutions:

The autonomous cotton harvesting system involves the combination of various subsystems that cut across the perception, motion planning and mechanical design. This section does not consider monolithic configurations of systems, but instead, takes a component-based approach wherein each decision point in the design is considered separately. The weights of the criteria are based on the Binary Dominance Matrix developed in Section 2.1 and the weights used in each decision matrix are based on the weights of the corresponding PDS criteria. The chosen components are then incorporated in the final system structure.

2.2.1 Solution Space Overview

There are 6 decision points in the autonomous cotton picking system, which are located in the software architecture, perception, control and mechanical domains. Table 2 is a summary of these decision points, their requirements based on the product design requirements (Section 2.1) and their effect on the overall system performance.

Table 2: Solution Space: Design Decision Points

Decision Point	Requirement Reference	Impact on System
Camera Placement	QL-03 (90% detection), SR-02 (mock field coverage)	Affects viewing angles, occlusion handling and inspection distance
Detection Method	QL-03 (90% detection), EN-01 (lighting robustness)	Affects detection reliability under varying conditions
Cluster Identification	QL-05 (90% pick rate), RL-02 (software reliability)	Determines tracking accuracy across multiple scan positions
Manipulator Configuration	SR-01 (520mm reach), WT-03 (tip deflection <10mm)	Affects workspace coverage and approach trajectory flexibility
Gripper Design	SR-04 (50-60mm opening), QL-05 (90% pick rate)	Determines harvest purity and foreign matter content
Reservoir Design	SR-03 (15×15×15cm), ER-03 (toolfree removal <15s)	Affects storage capacity and operator interaction

The next subsections offer the options of each decision point, compare them with the criteria and explain the choice of the solution.

2.2.2 Design Decision Analysis

Decision Point 1: Camera Placement

The first decision is on camera placement. The viewing angles, inspection distance and occlusion management are affected by the camera position. Two possibilities were considered in Table 3: (A) Eye-in-Hand: Camera attached to the end-effector which allows close-up inspection (about 35cm viewing range) and arbitrary positioning of the viewpoint; (B) Upper-Arm Mounted: Camera on a more proximate link which provides platform stability, but with reduced positioning flexibility.

Table 3: Camera Placement Decision Matrix

Criterion	Weight	PDS Ref.	Eye-in-Hand	Upper-Arm
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Close-up Inspection Capability	0.20	QL-03	5	2
Multi-angle Viewing	0.15	SR-02	5	3
Occlusion Handling	0.36	QL-03, EN-01	5	2
Platform Stability	0.20	QL-02	3	5
Calibration Simplicity	0.09	MA-03	3	4
Weighted Total	1.00		4,42	2,93

The close-up inspection and multi-viewpoint positioning criterion are important for reliable detection and measuring depth properly. The trade-off of stability is accepted and not a crucial problem in the case of controlled motion during scanning.

Decision Point 2: Detection Method

Cotton boll detection creates bounding boxes around the target on the captured images by the camera that defines the position of the target. Two alternatives were considered in Table 4: (A) Classical Computer Vision: Color segmentation (HSV thresholding), edge detection and blob analysis; computationally much more efficient, requires less sophisticated hardware but it is sensitive to changes in lighting and backgrounds; (B) Deep Learning (YOLO): CNN-based detection trained on labeled cotton datasets with excellent performance on detection under different lighting conditions and backgrounds, but requires more computational power.

Table 4: Detection Method Decision Matrix

Criterion	Weight	PDS Ref.	Classical CV	YOLO11
Robustness to Lighting Variation	0.18	EN-01	2	5
Detection Accuracy	0.23	QL-03	2	5
Handling Complex Backgrounds	0.41	QL-03, RL-01	2	5
Computational Cost	0.18	EN-02	5	3
Weighted Total	1.00		2,54	4,64

Selected: YOLO11

The deep learning is necessary to achieve the 90% detection accuracy when the field-like conditions are available. The lightning conditions, the background and the camera orientation affects the detectability of the cotton bolls significantly in Classical CV, while deep learning with YOLO neglects these negative effects for more precise detection in different conditions.

Classical CV Evaluation: Classical CV Evaluation: HSV (Hue-Saturation-Value) color space segmentation was experimented as the most promising classical method because white color of cotton can be separated through low saturation and high value threshold. Nevertheless, it was tested that about 50 percent false positive rate came about because of sky areas, lighting reflections, and background objects that have the same color properties (see Appendix Figure 15). This validates the need to have learned feature representations in order to have credible cotton detection.

Decision Point 3: Cluster Identification Strategy

In panoramic scanning, a single cluster can be identified in several perspectives. Two have been considered in Table 5: (A) Multi-Frame Tracking (ByteTrack): Video-based trackers that keep identity across frames through motion prediction [27][28]; continuous video with smooth motion of objects; (B) World Space 3D Clustering: Each detection is converted to 3D world coordinates with depth and TF transforms, and then grouped by spatial proximity regardless of camera motion.

Table 5: Cluster Identification Decision Matrix

Criterion	Weight	PDS Ref.	ByteTrack	World Space 3D
Stability Across Discrete Scan Positions	0.32	QL-05	2	5
ID Consistency	0.25	RL-02	3	5
Independence from Frame Rate	0.32	QL-04	2	5
Implementation Complexity	0.11	MF-03	4	3
Weighted Total	1.00		2,47	4,78

Selected: World-Space 3D Clustering

Experiments showed that with a large movement of camera between scan positions, ByteTrack ID becomes unstable. World-space clustering offers strong identification in relation to camera motion, with complete-linkage algorithm that gives tight clusters with no chain-linking artifacts.

Decision Point 4: Manipulator Configuration

The manipulator should be able to offer enough reach and allow approach paths that should not hit the plants. Two were compared in Table 6 (A) 4-DOF Robotic Arm: Base rotation, shoulder, elbow, wrist; simple positioning, though restricted approach angle range; (B) 6-DOF Robotic Arm: Full position and orientation control with gripper approach at arbitrary angles.

Table 6: Manipulator Configuration Decision Matrix

Criterion	Weight	PDS Ref.	4-DOF	6-DOF
Approach Trajectory Flexibility	0.29	QL-05	2	5
Workspace Coverage	0.23	SR-01, SR-02	3	4
Obstacle Avoidance Capability	0.35	SF-03	2	5
Mechanical Simplicity	0.13	MA-01	5	3
Weighted Total	1.00		2,62	4,51

Selected: Doosan M1013 6-DOF Robotic Arm

Extra DOF allow obstacle-free paths and the optimal placement of the gripper. The Doosan M1013 6-DOF arm reaches SR-01 satisfying 1300mm+.

Decision Point 5: Gripper Design

The gripper should be able to hold the cotton bolls with certainty and reduce the harvest contamination, while preserving the integrity of the cotton fibers. Two alternatives were considered in Table 7: (A) Parallel Jaw Gripper: Two opposing fingers with adjustable grip force, 50-60mm aperture to allow cotton clusters to be selected, (B) Vacuum/Suction: Negative pressure pickup to generate suction that would pick up cotton fibers.

Table 7: Gripper Design Decision Matrix

Criterion	Weight	PDS Ref.	Parallel Jaw	Vacumm
Foreign Matter Reduction	0.35	QL-05	5	2
Selective Grasping Capability	0.25	QL-05	5	2
Simplicity & Cost	0.20	MC-01, MF-01	5	4
3D Printability	0.10	MF-01	5	3
Energy Efficiency	0.10	EN-02	4	2
Weighted Total	1.00		4,85	2,45

Selected: Parallel Jaw Gripper (Robotiq Hand-E gripper)

Vacuum/suction systems are prone to picking up the surrounding debris (leaves, stems, dirt) and cotton fibers, which add to the content of foreign matter in the harvest. Parallel jaw grippers allow the selective pick of the target cotton boll to enhance harvest purity. Also, vacuum systems need constant air pump running that consumes more energy, and servo-driven fingers do not run until the grasp cycles. The 50-60mm aperture (SR-04) is possible using the Robotiq Hand-E gripper and our 3D-printable fingers that fit onto the gripper's jaws.

2.2.3 Final System Configuration

The chosen elements are combined (seen in Table 9) into a consistent system architecture with ROS2 as the communication platform, YOLO11 as the detection feed, and MoveIt2 as the planner generating collision-aware movements of the 6-DOF arm and the parallel gripper placing cotton into the removable reservoir.

Table 9: Selected Components Summary

Decision Point	Selected Option	Weighted Score	Key Justification
Camera Placement	Eye in Hand (Wrist)	4,42	Enables multi-angle inspection and close-up viewing
Detection Method	YOLO11	4,64	Robust detection under varying lighting conditions
Cluster Identification	World Space 3D Clustering	4,78	Stable across discrete scan positions, complete-linkage prevents chain-linking
Manipulator Configuration	6-DOF Doosan M1013	4,51	Flexible approach trajectories, sufficient reach
Gripper Design	Robotiq Hand-E Parallel Jaw Gripper	4,85	Reduces foreign matter, selective grasping, energy efficient

Also, an operator dashboard has been developed in the form of a web-based platform to meet the ergonomic requirements (ER 01, ER-02) mentioned in Section 2.1. The dashboard will offer convenient START/PAUSE/EMERGENCY controls, color-coded status displays and real-time visualization of ML confidence, which will allow flexible placement of the operators in the demonstrations. Section 2.3 will offer specifications of the design of each of the chosen components and their combination.

2.3. Detailed Design & Analysis

The integration of an heavy industrial manipulator onto the mobile bases introduces significant dynamic stability and tip over risks. In mobile manipulation the global Center of Mass (CoM) is dynamically altered by the continuous movement of the robotic arm. The tipping over risk is maximized when the robotic arm reaches its full extension. For the projects system with a Doosan M1013 robotic arm mounted on a Clearpath Husky mobile base, after an evaluation the initial positioning of the system demonstrated forward instability due the mass of the industrial manipulator. Because the system is designed for harvesting cotton bolls, the operational payload mass was determined to be negligible (0.0 kg). To prevent tipping, two mechanical solution strategies were investigated:

1. The addition of a rear counterweight to the mobile base.
2. The relocation of the robotic arm mounting position toward the center of the Husky platform.

Besides these mechanical solutions an operational constraint was already introduced based on the system's motion planning software. A "within reach" check is utilized which prevents the manipulator to operate in a risky way that can damage the system.

2.3.1. FORWARD STABILITY ANALYSIS

A baseline stability analysis was conducted on the original system configuration with a zero payload. The system consists of the Husky base, wheels, mount plate, rear reservoir, Doosan M1013 arm, and the end effector (Robotiq Hand-E and wrist camera). The total system mass was 96.62 kg. In the original design, the arm was mounted 0.20 m forward of the base center ($X = 0.20$ m) and reaches forward 1300 mm. The tipping moment is generated by the mass of the components hanging from the front axle ($X > 0.255$ m)

Extended Arm Mass:

- **Mass:** 34.23 kg
- **CoM Position:** $X = 0.83$ m (0.20 m *base* + 0.63 m forward extension center)
- **Lever Arm:** 0.83 m – 0.255 m = 0.575 m
- **Moment:** 34.23 kg $\times$ 9.81 m/s² $\times$ 0.575 m = 193.08 N

End-Effector Mass (Gripper + Camera):

- **Mass:** 0.39 kg
- **Position:** $X = 1.50$ m (0.20 m *base* + 1.30 m full reach)
- **Lever Arm:** 1.50 m – 0.255 m = 1.245 m
- **Moment:** 0.39 kg $\times$ 9.81 m/s² $\times$ 1.245 m = 4.76 N

Total Tipping Moment (M_{tip}):

$$M_{tip} = 193.08 \text{ Nm} + 4.76 \text{ Nm} = 197.84 \text{ N} \approx 197.9 \text{ N}$$

The Restoring Moment ($M_{restore}$)

The restoring moment acts as the anchor and is generated by the components sitting behind the front axle ($X < 0.255 \text{ m}$).

Component	Mass (kg)	Position X (m)	Lever Arm (m)	Moment (Nm)
Husky Body	50	0	0,26	125,08
Rear Wheels	4	-0,26	0,51	20,01
Reservoir Bin	2	-0,3	0,56	10,89
Mount Plate	2	0,2	0,06	1,08
Total	58			157,06

Total Restoring Moment ($M_{restore}$):

$$M_{restore} = 125.08 + 20.01 + 10.89 + 1.08 = 157.06 \text{ N} \approx 157.1 \text{ N}$$

Factor of Safety Calculation (FoS)

The Factor of Safety is the ratio of the anchoring force to the tipping force.

$$FoS = \frac{M_{restore}}{M_{tip}} = \frac{157.1 \text{ Nm}}{197.9 \text{ Nm}} = 0.7938$$

The Factor of Safety is nearly 0.79. Because this ratio is less than 1.0 that means the tipping forces overpower the restoring forces which confirm that the robot will physically tip forward over its front axle in this configuration.

Solution 1: Counterweight Addition Analysis

The first solution strategy was the addition of a dead counterweight to the rear of the mobile platform. A counterweight improves stability by generating an additional restoring moment around the pivot axis. To achieve a target FoS of 1.2, an additional restoring moment of 80.4 Nm was required. The counterweight was assumed placing at the rear at $X = -0.30 \text{ m}$ (a lever arm of 0.555 m from the front axle is created). The required counterweight mass (m_{cw}) was calculated as:

$$m_{cw} = \frac{80.4 \text{ Nm}}{9.81 \text{ m/s}^2 \times 0.555 \text{ m}} \approx 14.8 \text{ kg}$$

A mass of 15.0 kg was selected for the task.

Updated Center of Mass:

The total system mass increased to 111.62 kg.

$$X_{COM} = \frac{(96.62 \times 0.298) + (15.0 \times -0.30)}{111.62} = 0.217 \text{ m}$$

Stability Metrics:

- **Stability Margin:** $SM = 0.255 \text{ m} - 0.217 \text{ m} = +0.038 \text{ m}$

- **Factor of Safety:** The new restoring moment was calculated as 238.7 Nm

$$FoS = \frac{238.7 \text{ Nm}}{197.9 \text{ Nm}} = 1.21$$

While the counterweight solution prevented static tipping it introduced critical practical disadvantages that were affecting the system:

- **Increased Total Vehicle Mass:** The system mass increased by 15.5% to 111.62 kg.
- **Higher Energy Consumption:** More energy is required to accelerate the heavier mass.
- **Increased Drivetrain Stress:** The differential drive motors must produce higher torque to rotate the heavier platform.
- **Reduced Maneuverability:** The placement of 15 kg at the extreme rear significantly increases the mass moment of inertia (I_{zz}) around the vertical axis, causing a severe skid steer penalty.

Solution 2: Robotic Arm Shifting Analysis

The second mechanical solution was the relocation of the robotic arm mount to the exact center of the Husky platform ($X = 0.0\text{m}$, $Y = 0.0\text{m}$). By moving the heavy manipulator backward, the lever arm producing the tipping moment is drastically decreased without increasing system mass. Because this method solves the mass and maneuverability penalties associated with the counterweight, it is adopted as the foundational design.

Updated Tipping Over Analysis

The stability boundary is defined by the wheel contact patches of the Husky platform.

- Front Axle Tipping Line (X_{front}): +0.255 m
- Rear Axle Tipping Line (X_{rear}): -0.255 m
- Side Wheel Tipping Line (Y_{edge}): ± 0.2775 m
- Total System Mass: 96.62 kg

1) Forward Tipping Over Analysis

When the arm extends strictly forward to harvest a boll at $X = 1.30$ m, the extended arm's local CoM shifts to $X = 0.63$ m. New global center of mass X_{CoM} is calculated:

$$X_{CoM} = \frac{(60.0 \times 0.0) + (2.0 \times -0.30) + (34.23 \times 0.63) + (0.39 \times 1.30)}{96.62} = +0.222\text{m}$$

The CoM is safely maintained behind the 0.255 m front axle.

Factor of Safety (FoS):

$$M_{tipforward} = 34.23 \times 9.81 \times (0.63 - 0.255) + 0.39 \times 9.81 \times (1.30 - 0.255) = 129.92\text{Nm}$$

$$M_{restore_forward} = 60.0 \times 9.81 \times 0.255 + 2.0 \times 9.81 \times (0.255 - (-0.30)) = 160.98 \text{ Nm}$$

$$FoS_{forward} = 160.98/129.92 = 1.24$$

2) Lateral Tipping Over Analysis

When the arm swings 90° to the side to harvest a boll at Y = 1.30 m, the arm's CoM shifts laterally to $Y = 0.63$ m. When this happens the Global Center of Mass (Y_{CoM}) becomes:

$$Y_{CoM} = \frac{(62.0 \times 0.0) + (34.23 \times 0.63) + (0.39 \times 1.30)}{96.62} = 0.228 \text{ m}$$

This means the CoM is safely maintained inside the 0.2775 m side track.

Factor of Safety (FoS):

$$M_{tip_lateral} = 34.23 \times 9.81 \times (0.63 - 0.2775) + 0.39 \times 9.81 \times (1.30 - 0.2775) = 122.26 \text{ Nm}$$

$$M_{restore_lateral} = 62.0 \times 9.81 \times 0.2775 = 168.78 \text{ Nm}$$

$$FoS_{lateral} = 168.78/122.26 = 1.38$$

3) Backward Tipping Over Analysis

When the arm swings 180° and extends fully backward over the reservoir to X = -1.30m, the local CoM shifts to X = -0.63 m. the Global Center of Mass (X_{CoM}) is calculated by:

$$X_{CoM} = \frac{(2.0 \times -0.30) + (34.23 \times -0.63) + (0.39 \times -1.30)}{96.62} = -0.235 \text{ m}$$

This means the CoM is safely maintained inside the -0.255 m rear axle.

Factor of Safety (FoS):

$$M_{tip_rear} = 129.92 \text{ Nm (arm)} + (2.0 \times 9.81 \times (0.30 - 0.255)) \text{ Nm} = 130.80 \text{ Nm}$$

$$M_{restore_rear} = 60.0 \times 9.81 \times 0.255 = 150.09 \text{ Nm}$$

$$FoS_{rear} = 150.09/130.80 = 1.15$$

The calculations demonstrate that the lateral Factor of Safety (1.38) is significantly higher than the forward Factor of Safety (1.24). This is caused entirely by the rectangular shape of the Husky chassis support polygon. The distance from the center of the robot to the wheels is the leverage the chassis has to keep the robot grounded.

- The longitudinal pivot (wheelbase) is located at 0.255 m.
- The lateral pivot (track width) is located at 0.2775 m.

Because the side wheels are positioned further out, the 60 kg steel chassis of the Husky possesses a longer lever arm when opposing a lateral tip. This generates a restoring force of 168.78 Nm laterally, compared to only 160.98 Nm longitudinally.

	Original Design (Forward)	Counterweight (Forward)	Shifted Design (Forward)	Shifted Design (Lateral)	Shifted Design (Backward)
Total Mass	96.62 kg	111.62 kg	96.62 kg	96.62 kg	96.62 kg
Stability Margin	-0.043 m	+0.038 m	+0.033 m	+0.049 m	+0.020 m

Factor of Safety	0.79	1,21	1,24	1,38	1,15
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2.3.2. SINGULARITY ANALYSIS

The robotic harvesting platform has a Doosan M1013 6 DOF industrial manipulator mounted on a Clearpath Husky mobile base. The manipulator is required to reach cotton bolls during harvesting operations located at various heights and distances. As the arm approaches its workspace boundaries, kinematic singularities may occur that leading to reduced controllability, poor trajectory tracking, excessive joint velocities and potentially unsafe operating conditions. To evaluate these risks, a kinematic analysis was done using the Denavit Hartenberg (D-H) formulation and Jacobian based singularity analysis. The resulting singularity conditions were then used to develop avoidance strategies utilizing both manipulator motion and mobile base redundancy. The Doosan M1013 is modeled as a six revolute joint serial manipulator with a spherical wrist. The Denavit-Hartenberg Parameters used in the analysis are presented in Table 1 where q are joint variables a_2 and a_3 are primary arm link lengths.

Table 2: D-H Parameters of the Doosan M1013

Joint (i)	θ_i	d_i (m)	a_i (m)	α_i (rad)
1	q_1	0.152	0	$\pi/2$
2	q_2	0	0.620	0
3	q_3	0	0.559	$\pi/2$
4	q_4	0.121	0	$-\pi/2$
5	q_5	0	0	$\pi/2$
6	q_6	0.140	0	0

The forward kinematics are obtained by multiplying the individual homogeneous transformation matrices which provide the complete position and orientation of the end effector:

$$T_0^6 = T_0^1 T_1^2 T_2^3 T_3^4 T_4^5 T_5^6$$

Geometric Jacobian Formulation

The manipulator Jacobian relates joint velocities to Cartesian end effector velocities according to

$$v = J(q)\dot{q}$$

where

$$v = [v_x \quad v_y \quad v_z \quad \omega_x \quad \omega_y \quad \omega_z]^T$$

and

$$\dot{q} = [\dot{q}_1 \quad \dot{q}_2 \quad \dot{q}_3 \quad \dot{q}_4 \quad \dot{q}_5 \quad \dot{q}_6]^T$$

The geometric Jacobian is expressed as

$$J(q) = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix}$$

where the upper three rows represent linear velocity contributions and the lower three rows represent angular velocity contributions. For a revolute joint, the i -th Jacobian column is calculated as

$$J_i = \begin{bmatrix} z_{i-1} \times (P_6 - P_{i-1}) \\ z_{i-1} \end{bmatrix}$$

Where z_{i-1} is the axis of rotation of Joint i , P_6 is the TCP position, P_{i-1} is the origin of Frame $i-1$. The complete Jacobian becomes:

$$J(q) = \begin{bmatrix} z_0 \times (P_6 - P_0) & z_1 \times (P_6 - P_1) & z_2 \times (P_6 - P_2) & z_3 \times (P_6 - P_3) & z_4 \times (P_6 - P_4) & z_5 \times (P_6 - P_5) \\ z_0 & z_1 & z_2 & z_3 & z_4 & z_5 \end{bmatrix}$$

For the Jacobian based singularity analysis, a manipulator singularity occurs whenever

$$\det(J) = 0$$

At this condition, the robot loses one or more independent motion directions and the inverse kinematics become badly conditioned. Because the Doosan M1013 possesses a spherical wrist, the singularity analysis can be simplified through kinematic decoupling.

Wrist-Center Decoupling

The rotational axes of Joints 4, 5 and 6 intersect at a common wrist center. By shifting the reference point from the TCP to the wrist center, the Jacobian assumes a block upper-triangular structure:

$$J = \begin{bmatrix} J_{11} & 0 \\ J_{21} & J_{22} \end{bmatrix}$$

Where J_{11} describes arm positioning J_{22} describes wrist orientation so the determinant becomes:

$$\det(J) = \det(J_{11}) \det(J_{22})$$

Thus singularities can be analyzed separately for the arm and wrist subsystems. The Jacobian is evaluated at the wrist center, which eliminates the influence of joints 4–6 on the translational subJacobian and enables block diagonal decoupling

Wrist Singularity

When we write out the math for the robot's wrist, the result is

$$\det(J_{22}) = -\sin(q_5)$$

Setting the determinant equal to zero gives

$$\sin(q_5) = 0$$

which results in

$$q_5 = 0^\circ \quad \text{or} \quad q_5 = \pm 180^\circ$$

At this configuration, the rotational axes of Joints 4 and 6 become perfectly aligned. As a result, the manipulator loses one rotational degree of freedom and certain orientations become mathematically impossible to achieve independently. This condition is known as the wrist singularity.

Manipulability Analysis

To quantify proximity to singular configurations, Yoshikawa's manipulability index is employed:

$$w = \sqrt{\det(JJ^T)}$$

The index provides a scalar measure of movement score, the freedom to move. Large values of w indicate high freedom to move small values indicate lower controllability. $w=0$ means a singularity configuration. After the Jacobian based singularity analysis it is seen that wrist singularities occur when $q_5 = 0^\circ$ or $q_5 = \pm 180^\circ$. Manipulability decreases rapidly as these configurations are approached. The redundancy from the mobile base can be used to avoid these singularity regions by repositioning the vehicle during operation. These strategies provides an effective framework for the safe and reliable autonomous cotton harvesting operation.

2.3.3. DYNAMIC ANALYSIS

While static tipping over and kinematic singularity analyses confirm the system's viability during stationary harvesting, navigating through agricultural fields introduces severe dynamic loads. The electromechanical capability of the Husky base to transport the 96.62 kg system across soft agricultural soil at operational speeds is evaluated. The analysis investigates straight line tractive effort, skid steer turning torque limits and the inertial impacts of maintaining a functional "Home Position" during navigation.

System Parameters and Environmental Constraints

Vehicle Parameters (Husky A200):

- Total System Mass (m): 96.62 kg
- Wheel Radius (r): 0.165 m
- Track Width (B): 0.555 m
- Wheelbase (L): 0.510 m
- Continuous Combined Motor Torque: ≈ 44.0 Nm

Environmental and Operational Parameters:

- Terrain Type: Agricultural Soil
- Coefficient of Lateral Friction (μ): ≈ 0.6
- Coefficient of Rolling Resistance (C_{rr}): ≈ 0.15
- Max Navigation Velocity (v_{max}): 1.0 m/s
- Max Navigation Acceleration (a): 0.5 m/s²

To navigate the aisles in the field simulation efficiently, the Husky's drivetrain must overcome both the inertia of the heavy manipulator system and the high rolling resistance associated with soft, yielding soil.

Rolling Resistance Force (F_r):

The continuous force required to push the 96.62 kg mass through soft dirt without accelerating.

$$F_r = C_{rr} \cdot m \cdot g = 0.15 \cdot 96.62 \text{ kg} \cdot 9.81 \text{ m/s}^2 = 142.18 \text{ N}$$

Acceleration Force (F_a):

The force required to reach the maximum velocity of 1.0 m/s.

$$F_a = m \cdot a = 96.62 \text{ kg} \cdot 0.5 \text{ m/s}^2 = 48.31 \text{ N}$$

Total Required Drive Torque (τ_{drive}):

The total linear force required at the wheel contact patches is 190.49 N. Converting this to rotational torque at the axles:

$$\tau_{drive} = (F_r + F_a) \cdot r = 190.49 \text{ N} \cdot 0.165 \text{ m} = 31.43 \text{ Nm}$$

The required torque of 31.43 Nm is within the Husky's continuous limit of 44.0 Nm. This means the rover is fully capable of driving straight and accelerating on soil without stalling.

The Husky A200 uses a differential drive (skid-steer) mechanism. Turning requires the left and right wheels to drive in opposite directions, this means forcefully skidding the rubber tires laterally across the high friction soil.

Resisting Turning Moment (M_{res}):

The soil generates a massive friction torque fighting against the robot's attempt to do a zero radius turn (turning without moving).

$$M_{res} = \frac{\mu \cdot m \cdot g \cdot L}{2} = \frac{0.6 \cdot 96.62 \text{ kg} \cdot 9.81 \text{ m/s}^2 \cdot 0.510 \text{ m}}{2} = 145.03 \text{ Nm}$$

Required Skid-Steer Motor Torque (τ_{turn}):

To generate 145.03 Nm of steering moment, the motors must apply a differential lateral force (F_{skid}) acting across the track width (B).

$$F_{skid} = \frac{M_{res}}{B} = \frac{145.03 \text{ Nm}}{0.555 \text{ m}} = 261.32 \text{ N}$$

$$\tau_{turn} = F_{skid} \cdot r = 261.32 \text{ N} \cdot 0.165 \text{ m} = 43.12 \text{ Nm}$$

A turning torque of 43.12 Nm is critically close to the Husky's absolute continuous limit of 44.0 Nm (98% of capability). Attempting a zero radius turn on soft soil poses a high risk of triggering motor driver overcurrent protection faults, which would stall the system in the field.

Dynamic Stability and the "Home Position" Advantage

To protect the system during transit, the manipulator is maintained in a functional "Home Position". This operational constraint provides two major dynamic advantages:

1. **Neutralization of the Pendulum Effect:** If left extended, the 34.23 kg arm mass mounted on the 0.30 m deck would create a high global Center of Mass ($Z_{CoM} \approx 0.45 \text{ m}$). Navigating uneven agricultural terrain would induce violent pitching (the "wobble effect")
2. **Optimized Normal Force Distribution:** By folding the arm over the exact center of the vehicle ($X=0, Y=0$), the normal force (weight) is distributed equally across all four tires. This prevents the front tires from digging into the soft soil, which would otherwise spike the lateral friction coefficient (μ) and guarantee a motor stall during turns.

In conclusion of the dynamic analysis The Husky base possesses the electromechanical capacity to transport the Doosan M1013 manipulator across agricultural terrain, provided strict operational limits are enforced within the ROS 2 navigation stack. By implementing a set of motion planning rules like the system only being able to move when the robot arm is in its "Home Position".

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2.3.3 Vision System

The vision system combines an RGB-D camera for color and depth acquisition with a deep learning detector for cotton boll recognition. In the current stage of the project, the “eye in hand” camera is simulated using Gazebo's `rgbd_camera` sensor plugin, configured to match the Intel RealSense D435i camera planned for hardware (see Appendix Table 24 for full specifications). The camera follows the standard pinhole projection model seen in the Figure 13. The intrinsic matrix K encapsulates internal parameters:

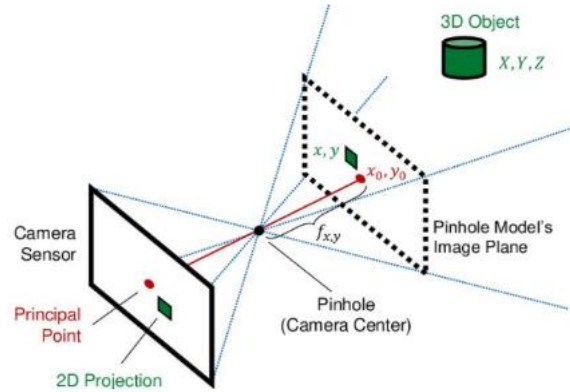


Figure 13: Pinhole Camera Model [35]

Equation 1: Camera Intrinsic Matrix:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 277 & 0 & 320 \\ 0 & 277 & 240 \\ 0 & 0 & 1 \end{bmatrix}$$

Where $f_x, f_y = 277$ pixels (focal length derived from FOV: $f_x = 320/\tan(45^\circ) \approx 277$), and $c_x=320, c_y=240$ (principal point at image center).

Equation 2: Perspective Projection (3D to 2D)

$$\begin{aligned} u &= f_x \times (X / Z) + c_x \\ v &= f_y \times (Y / Z) + c_y \end{aligned}$$

YOLO (You Only Look Once) object detection model is a single stage object detector consisting of a CSPDarknet backbone for feature extraction, a PANet neck for multi scale feature fusion and a detection head that predicts bounding boxes with class probabilities in one forward pass [12][22]. This architecture enables real time inference by processing the entire image simultaneously rather than using region proposals.

2.3.4 Dataset Collection and Annotation

The latest dataset was prepared from the updated Roboflow export used for the final YOLO11 training run. It combines real cotton images with synthetic and 3D-rendered cotton scenes so that the detector sees mature white bolls, unripe bolls in different viewpoints, lighting variation, distance changes, and partial occlusion. The final split contains 603 training images, 57 validation images, and 29 held-out test images.

Annotations were exported in YOLO format with two detection classes: cotton_boll and unripe-cotton. Cluster-level behavior is handled later in the ROS2 pipeline by grouping individual boll detections, rather than by training a separate cluster class.

Model Training

RoboCot now uses a YOLO11n detection model from Ultralytics. The model was trained for 80 epochs at 512 x 512 image size with batch size 4 on the local NVIDIA RTX 3050 GPU. The resulting best.pt weights are packaged for ROS2 deployment together with the data.yaml class list, training arguments, validation results, and metric plots. Based on the F1-confidence curve, the recommended starting confidence threshold is approximately 0.54; this can be increased toward 0.60-0.70 if the robot produces too many false positive pick attempts.

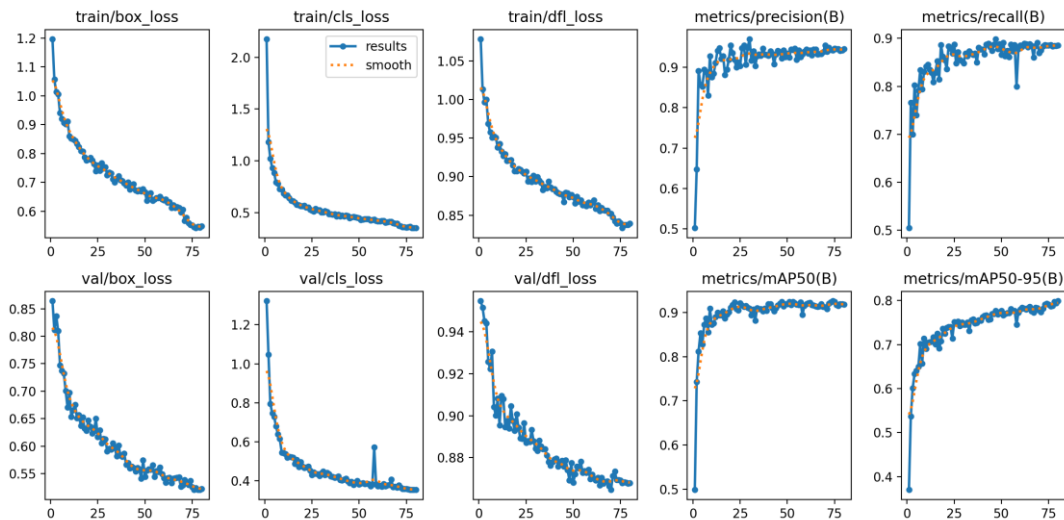


Figure 14: Training and validation curves for the latest YOLO11n training run.

The training curves in Figure 14 show stable convergence. Training and validation losses decrease through the run, while mAP50 rises quickly and remains stable. The best validation

row by mAP50 occurred at epoch 74 with precision 0.943, recall 0.885, mAP50 0.926, and mAP50-95 0.794. The final epoch reached mAP50-95 0.800, showing improved localization quality compared with the previous baseline model.

Detection Accuracy

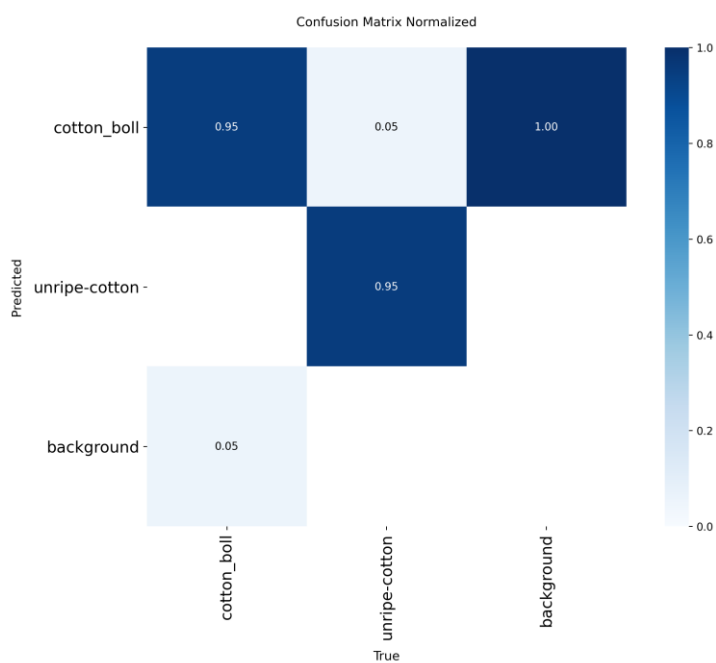


Figure 15: Normalized confusion matrix for the held-out test set.

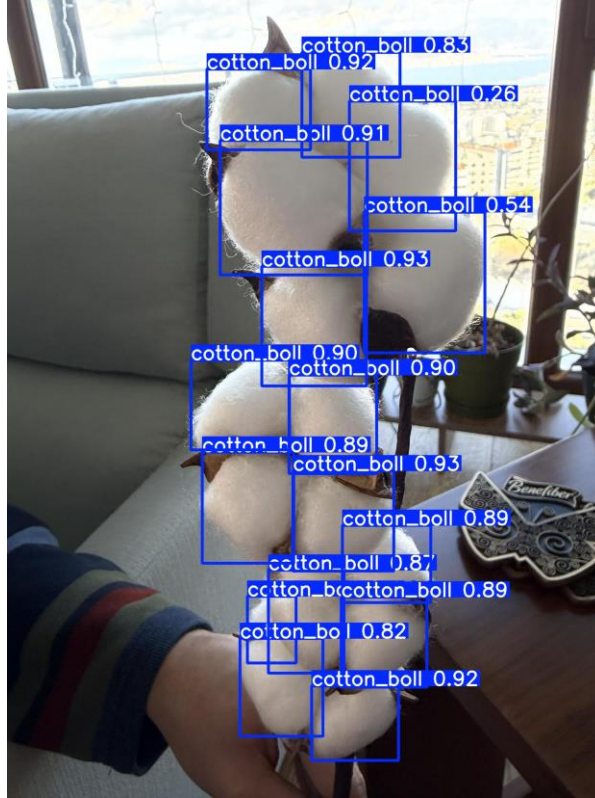


Figure 17: Sample detection on real cotton.

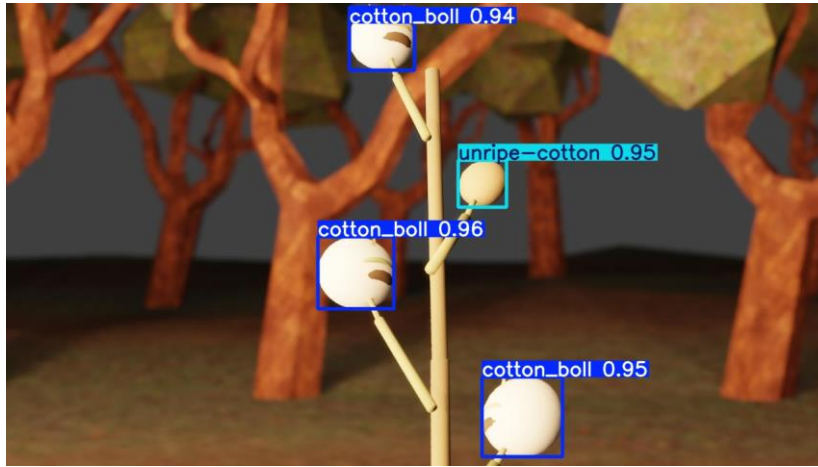


Figure 18: Detection on a cluster containing unripe cotton bolls.

Table 12: YOLO detection results by class on the held-out test split.

Class	Precision	Recall	mAP50	mAP50-95
All classes	0.956	0.971	0.972	0.806
cotton_boll	0.913	0.946	0.950	0.811
unripe-cotton	1.000	0.995	0.995	0.801

The held-out test results show strong detection performance, with all-class precision of 0.956, recall of 0.971, mAP50 of 0.972, and mAP50-95 of 0.806. The normalized confusion matrix indicates that both cotton_boll and unripe-cotton are recognized with approximately 95% recall at the plotted operating point. The main remaining errors are cotton_boll false positives on background and a small number of missed cotton_boll instances, so real camera testing should still be used before treating the model as field-final.

3. MANUFACTURING PROCESS

3.1 Material Selection

Since the RoboCot system primarily uses commercial main equipments like the Husky Mobile Base as the rover and Doosan M1013 robotic arm as the manipulator, also as the system has transitioned to a fully simulation environment rather than a physical prototype the material selection is mostly focused on the custom integration components. These custom parts include the camera mount bracket, the two gripper finger extensions and the reservoir bin for cotton bolls mounted on the deck of the rover base. The materials for these components were selected to optimize the structural rigidity, overall mass and ease of manufacturing.

3.1.1 ABS for 3D Printed Components

Acrylonitrile Butadiene Styrene (ABS) was selected for the custom 3D printed components, which consist of the camera mount bracket and the two gripper finger extensions. ABS was chosen over standard Polylactic Acid (PLA) because it offers superior impact resistance, toughness and durability. These mechanical characteristics are needed to withstand the repeated cyclic loading that occurs during the continuous picking and dropping operations of the project's end effector, the gripper of the Doosan M1013 robotic arm. Beside these ABS has a low density which guarantees that the custom extensions for the gripper and the camera mount which is also attached to the grippers top, adds minimal mass to the robotic arm. Keeping the weight low is critical for preserving the payload capacity of the Doosan M1013 and ensuring that the dynamic movements of the arm are not obstructed. As these components are placed on the end of the robotic manipulator, the weight factor is also very critical for the tipping over analyses as they have the potential to create a significant moment when the arm is in the fully reached out position. The inherent stiffness of ABS is necessary for the camera mount specifically to prevent the Intel Realsense camera from vibrating or shifting during the rapid pan and tilt scanning sweeps. This stiffness ensures that the spatial relationship between the camera and the wrist remains exactly constant, which is required for accurate 3D localization and it is one of the most important processes for the entire system. These components were actually printed and tested on the actual Doosan M1013 robotic arm in Kandilli Campus as they were already designed and manufactured before the transition to the fully simulation scope, they were exact matches and successful components for the robotic manipulator.

3.1.2 Corrugated Polypropylene for The Reservoir

Corrugated Polypropylene plastic, commonly referred to as Coroplast was selected for the collection reservoir mounted on the deck. The primary design constraint for the reservoir was the need to establish a large, distinct physical volume without adding excess mass to the Husky A200 platform. Adding heavy materials to the upper deck would raise the system's center of gravity and negatively impact the tipping over stability of the mobile base during aisle navigation. Corrugated polypropylene is built with internal plastic walls that provide strength similar to structural beams (series of I beams). This geometry provides significant compressive strength and resistance to bending while maintaining an extremely low effective density. Since a single mature cotton boll weighs only 3 to 5 grams, the total expected payload force exerted on the reservoir walls is minimal. The rigid nature of the hard plastic ensures that the reservoir maintains its precise box shape during the sudden accelerations, braking and turning of the mobile base. Consequently, the material successfully acts as a solid, self supporting structure for the physical prototype while contributing nearly zero weight to the system's payload budget. As this component was shifted from a flip top bin after the addition of the rover base to the

project it was not physically manufactured but designed to be most accurate to a future physical prototype.

3.2 Selection of Manufacturing Processes

3.2.1 Fused Deposition Modeling (FDM) 3D Printing

Fused Deposition Modeling (FDM) 3D printing was selected to manufacture the custom ABS components the camera mount and the gripper finger extensions. The Bambu Lab X1 Carbon 3D printer in the North Campus of Boğaziçi University was used for the manufacturing of the components. This specific FDM system was chosen because the actively controlled high temperature enclosure ensures excellent layer adhesion and minimizes thermal warpin which is critical when processing ABS for the precise manufacturing of the components specifically the camera mount with precise screw holes designed to unite the camera and the Doosan robotic arm. Consequently, the printed components achieve enhanced structural integrity and high dimensional accuracy. This precision was essential for the camera mount bracket to strictly maintain the spatial offsets required for the vision pipeline. Additive manufacturing provided the flexibility to adjust and minimize the total mass added to the robotic arm's payload [X].

3.2.2. Sheet Cutting and Folding

For the fabrication of the collection reservoir mounted on the deck, 2D sheet cutting and folding techniques were selected. Since corrugated polypropylene is acquired as premanufactured flat sheets, complex forming or molding processes are avoided. To construct the reservoir, the material is needed to be precisely cut to the required dimensional profile. Following this, fold lines are scored parallel to the material's internal fluted structure. As mentioned before the actual physical reservoir was not manufactured due to the process scope changed but it was placed inside the simulation environment in Gazebo. Sheet cutting and folding technique allows the 2D sheets to be rapidly folded into a rigid, three-dimensional box volume. This specific manufacturing method was chosen because it enables the efficient construction of a large, highly rigid physical boundary for the cotton bolls. Most importantly, it achieves this while keeping the overall mass near zero meaning preserving the mobile platform's center of gravity and tipping over stability during operation.

3.4 Cost Analysis

This section summarizes the expected procurement costs of the required hardware components and also the testing environment which is the mock field for the project if the simulation environment would be physically actualized in the future. 3D Printed materials were manufactured on the North Campus of the university so no additional charges were received. Other items were already existing materials in the Teknopark of the Kandilli Campus in Professor Evren Samur's Lab meaning no extra budgeting were necessary to acquire them also. A market research was done and the suppliers that best fit our and the University's budget and delivery time constraints are chosen for ordering in the first semester for ME429 and Three Pro forma invoices were received from these but not procured as the materials shifted to the ones listed in Table below. The goal is to understand the total budget which is the combination of our own budget and the University's budget and the cost drivers of the project as a whole.

Table 3: Materials Covered by the University

Suppliers	Items	Quantity	Unit Price (Excl. VAT)	Total (Incl. VAT)
Boğaziçi University	Jetson AGX Xavier 64 GB	1	45.917,12 TRY	55.100,54 TRY
Boğaziçi University	AC8265 Wireless Network Adapter	1	897,30 TRY	1076,76 TRY
Boğaziçi University	Intel RealSense Depth Camera D435	1	19.612,80 TRY	24.516,00 TRY
Boğaziçi University	Doosan M1013 Robotic Arm + Gripper	1	47.500,00 USD	47.500,00 USD
Boğaziçi University	3D Printed Camera Mount	1	300,00 TRY	300,00 TRY
Boğaziçi University	3D Printed Gripper Extensions	2	75,00 TRY	150,00 TRY
Total Cost			2.177.038,00 TRY	2.192.742,00 TRY

Table below shows the items covered and planned to be covered (PP Plastic Sheet (Reservoir)) if the project was physically manufactured for the real life prototype from our own budgets.

Table 4: Materials Covered from Our Budget

Item	Quantity	Unit Price (incl. VAT)	Total (incl. VAT)
Cotton branch with bolls	30	75,00 TRY	2.250,00 TRY
Strafor Foam	300mm x 100mm x 30 mm	300,00 TRY	900,00 TRY
PP Plastic Sheet (Reservoir)	2 x (600mm x 600 mm)	192,00 TRY	384,00 TRY
Total Cost			3.534,00 TRY

4. PROJECT MANAGEMENT

The task distribution was carefully conducted for a well balanced project flow and the team members were under equal responsibilities in various tasks they were most confident in. In **Table** , the subtasks and their responsible group members in ME492 can be seen.

Table 5: ME492 Task Distribution

SUBTASKS	RESPONSIBLE GROUP MEMBER
Patent and Online Research	Collective
System Requirements & Constraints Definition	Mehmet Baran Erden
Component & Material Selection	Collective
Gripper Extension Design	Deniz Gerdan
Camera Mount Design	Mehmet Baran Erden
Gazebo Simulations	Ayhan Oruç
3D Printing	Mehmet Baran Erden
Examining and Understanding RoboCob	Collective
Extended Data Collection & Labeling	Deniz Gerdan
Robot Simulation & Vision-to-Motion Pipeline Development	Ayhan Oruç
ROS Learning	Collective
ROS2 Software Architecture & System Integration	Ayhan Oruç
Developing & Training AI Vision Model	Deniz Gerdan
Procurement Proposal Preparation	Mehmet Baran Erden
MT & Final Presentations	Collective

Resources

The resources for "Robocot" were initially selected to build a physical lab scale prototype. The original planned hardware stack was centered around an NVIDIA Jetson Orin NX for edge computing, a ZED X Mini stereo camera for perception and a Doosan Robotics six degree of freedom robotic arm for manipulation. Due to hardware limitations about the robot arm's usability the project scope was changed to a high fidelity digital simulation. This shift allowed the testing of the system's logic and kinematics. The finalized resources driving the project are now centered on a robust simulation framework built using ROS2 and Gazebo. This software stack provides the high fidelity physics engine and node based communication architecture necessary to replicate real world kinematics and sensor interactions. In place of the physical hardware, the main manipulator is a fully articulated 6-DOF robotic arm modeled by the Unified Robot Description Format (URDF) of the actual Doosan Robotics robot arm within the Gazebo environment. This setup allows for rapid, repeatable pick and place testing while strictly respecting simulated physical limits. To replicate the intended hardware capabilities, the physical stereo camera was replaced by a simulated depth camera sensor attached to the digital twin. This virtual sensor captures high resolution synthetic RGB and depth data, establishing a clear vision system to feed the YOLO11 machine learning model for target localization and approach planning. Finally, the high level vision inference and low level deterministic logic originally intended for the Jetson and Arduino hardware are handled by virtualized ROS2 controllers running on the host workstation. This digital architecture manages path planning, coordination and simulated safety triggers, ensuring that the developed code remains highly transferable for future physical implementations.

Gantt Charts

The Gantt Chart in **Figure 23** shows the project flow acquired during ME429 last semester and the Gantt Chart in **Figure 24** shows the work flow in for the project this semester in the ME492 course. It was not possible to fully actualize everything initially intended because of the problems caused by the unusability of the robotic arm Kandilli Campus, but the project scope were changed to a fully simulation environment and the testing processes were conducted on the high fidelity simulation. Even it was not possible to fully track the planned timeline with issues like this, with hard work conducted after the shift in the project scope the planned task were generally in line with the intended times and all tasks other than the physical building of the prototype were fully completed at the end of the ME492 course.

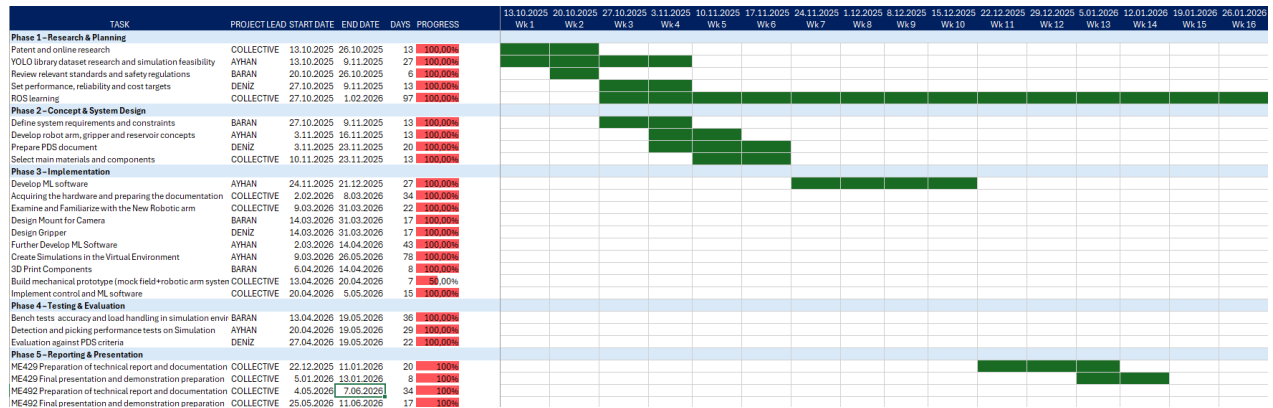


Figure 11: Gantt Chart of ME429

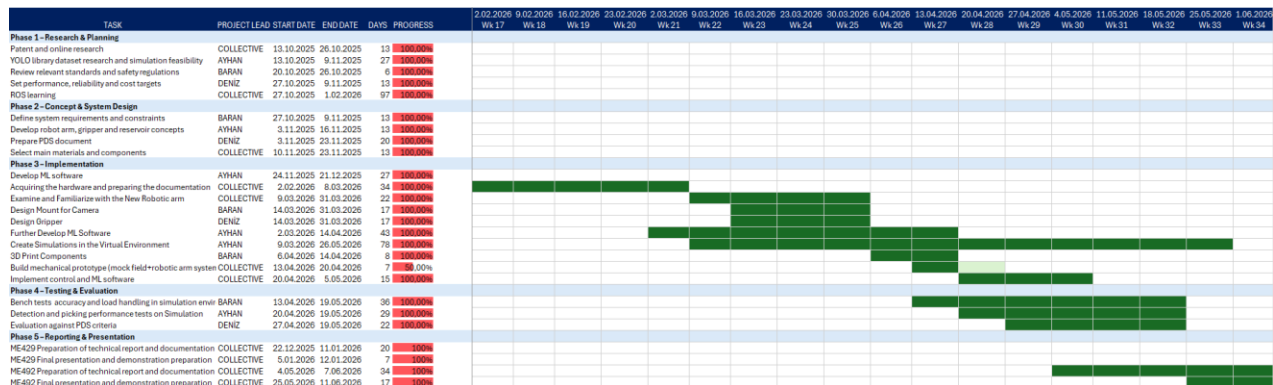


Figure 12 : Gantt Chart of ME492

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Figure 8: <https://www.agricultural-robotics.com/robot/swarmbot-5>

Figure 9:

Figure 10:

Figure 11:

